

Students used three classification tests to study the reactivity of alcohol, aldehyde, and ketone functional groups in organic compounds. The organic compounds provided to the students are given in Table 1.

Table 1 Organic Compounds Tested

Compound	Structural Formula
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$
C	$(\text{CH}_3)_3\text{COH}$
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$

First, the students performed the Lucas test by mixing a few drops of each compound with 2 mL of the Lucas reagent (ZnCl_2 and HCl). This reagent is used to test for secondary and tertiary alcohols. A positive test result is indicated by a cloudy reaction mixture upon addition of the Lucas reagent due to the formation of an insoluble compound by an $\text{S}_{\text{N}}1$ reaction with a secondary or tertiary alcohol (Reaction 1).

R

—

OI

Reaction 1

Next, a test for alcohols and aldehydes was performed by the addition of a drop of chromic acid reagent (CrO_3 dissolved in aqueous H_2SO_4) into samples of the organic compounds. The disappearance of the orange color from chromic acid and formation of a blue-green suspension is indicative of a positive result for this test. Compound A was found to be an example of a positive result, as shown in Reaction 2.

CH_3

Reaction 2

Finally, samples of the compounds were also dissolved in water, followed by the addition of NaOH and iodine. This test is known as the iodoform test, which is used to identify methyl ketones and secondary alcohols. A positive result is indicated by the formation of iodoform (CHI_3), a yellow precipitate (Reaction 3).

R

—

C

Reaction 3

Table 2 Selected Observations from the Classification Tests

Compound	Formula	Lucas test	Chromic acid test	Iodoform test
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$	Cloudy mixture	Blue-green suspension	Yellow precipitate
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$		Blue-green suspension	No precipitate
C	$(\text{CH}_3)_3\text{COH}$	Cloudy mixture		
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$	No reaction		
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$		Orange reaction mixture	Yellow precipitate

Based on the information in the passage, which compound will react most rapidly with the Lucas reagent?

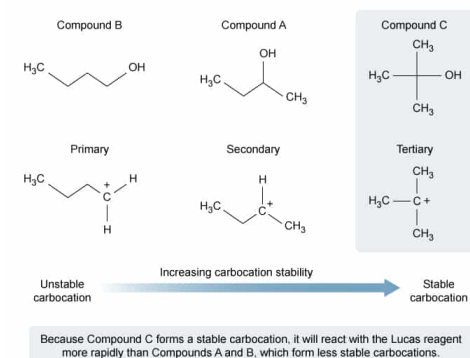
- ☐ A. Compound A [30%]
☐ B. Compound B [7%]
☒ C. Compound C [57%]
☐ D. Each compound will react at the same rate [4%]

Correct

57% answered correctly

Explanation:

Substituted alcohol reactivity in the Lucas test



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Substitution reactions can occur in one of two ways: an S_N1 or S_N2 mechanism. An S_N1 reaction occurs in **two steps** whereas an S_N2 reaction occurs in a single step. The first step of an S_N1 reaction involves formation of a **carbocation intermediate** when the bond between the leaving group and the adjoining carbon is broken. In the second step, the nucleophile adds to the carbocation, forming the product of the reaction.

The first step is the slowest step (**rate-determining step**). **Substrates** that form a more **stable carbocation** react **faster** in an S_N1 reaction than those that form a less stable carbocation.

The passage states that the **Lucas test** occurs through an S_N1 mechanism. Because Compound C is a **tertiary alcohol**, it can form a more stable carbocation than Compounds A and B, which are a secondary and primary alcohol, respectively. Therefore, Compound C will react the most rapidly with the Lucas reagent.

(Choice A) Compound A forms a secondary carbocation, which is less stable than the tertiary carbocation formed by a tertiary alcohol. Therefore, Compound A reacts more slowly than Compound C in the Lucas test.

(Choice B) Compound B is a primary alcohol and would form a primary carbocation in the S_N1 reaction if it reacted with the Lucas reagent. Because primary carbocations are not stable, it is unfavorable for Compound B to undergo an S_N1 reaction.

(Choice D) Compounds A and B react with the Lucas reagent at different rates due to carbocation stability. Compound B does not react with the Lucas reagent because it would form an unstable primary carbocation.

Educational objective:

An S_N1 reaction occurs in two steps. First, a carbocation intermediate is formed when the bond between the leaving group and adjoining carbon atom is broken; next, the nucleophile adds to the carbocation to form the reaction product. Substrates that form a more stable carbocation react faster than substrates that form a less stable carbocation.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401924

System:
Functional Groups and Their Reactions

Students used three classification tests to study the reactivity of alcohol, aldehyde, and ketone functional groups in organic compounds. The organic compounds provided to the students are given in Table 1.

Table 1 Organic Compounds Tested

Compound	Structural Formula
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$
C	$(\text{CH}_3)_3\text{COH}$
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$

First, the students performed the Lucas test by mixing a few drops of each compound with 2 mL of the Lucas reagent (ZnCl_2 and HCl). This reagent is used to test for secondary and tertiary alcohols. A positive test result is indicated by a cloudy reaction mixture upon addition of the Lucas reagent due to the formation of an insoluble compound by an $\text{S}_{\text{N}}1$ reaction with a secondary or tertiary alcohol (Reaction 1).

R

—

OH

Reaction 1

Next, a test for alcohols and aldehydes was performed by the addition of a drop of chromic acid reagent (CrO_3 dissolved in aqueous H_2SO_4) into samples of the organic compounds. The disappearance of the orange color from chromic acid and formation of a blue-green suspension is indicative of a positive result for this test. Compound A was found to be an example of a positive result, as shown in Reaction 2.

CH_3

Reaction 2

Finally, samples of the compounds were also dissolved in water, followed by the addition of NaOH and iodine. This test is known as the iodoform test, which is used to identify methyl ketones and secondary alcohols. A positive result is indicated by the formation of iodoform (CHI_3), a yellow precipitate (Reaction 3).

R

—

C

Reaction 3

Table 2 Selected Observations from the Classification Tests

Compound	Formula	Lucas test	Chromic acid test	Iodoform test
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$	Cloudy mixture	Blue-green suspension	Yellow precipitate
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$		Blue-green suspension	No precipitate
C	$(\text{CH}_3)_3\text{COH}$	Cloudy mixture		
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$	No reaction		
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$		Orange reaction mixture	Yellow precipitate

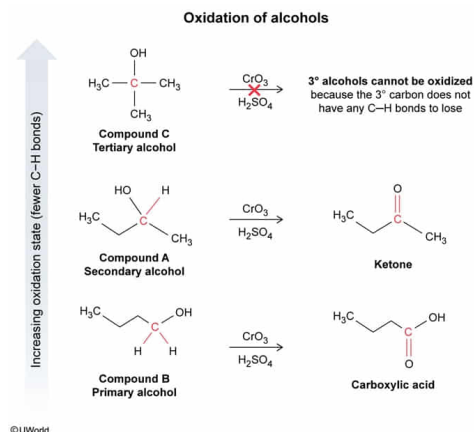
If the chromic acid reagent is added to Compound C, no reaction occurs because:

- ☐ A. a tertiary alcohol cannot be reduced. [13%]
- ☒ B. a tertiary alcohol cannot be oxidized. [80%]
- ☐ C. an aldehyde cannot be oxidized. [4%]
- ☐ D. an aldehyde cannot be reduced. [1%]

Correct

80% answered correctly

Explanation:



Oxidation of an **organic molecule** requires a **decrease** in the number of **C–H bonds** and an **increase** in the number of **C–O bonds**. **Primary and secondary alcohols** can be oxidized to form carboxylic acids and

ketones, respectively, using various **oxidizing agents** such as **chromic acid**. However, the tertiary carbon of a tertiary alcohol does not have any C–H bonds to lose; therefore, its number of bonds to oxygen cannot be increased. If the number of bonds to oxygen were increased without losing a bond to hydrogen, the carbon atom would have five bonds exceeding the maximum number possible (four bonds).

Reaction 2 shows that $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$ (Compound A), a secondary alcohol, gives a positive result for the chromic acid test and that it is converted into $\text{CH}_3\text{CH}_2\text{C}(\text{O})\text{CH}_3$, a ketone, in the process. As a result, an increase in the number of C–O bonds is achieved in the reaction indicating that the chromic acid test is an oxidation. Because $(\text{CH}_3)_3\text{COH}$ (Compound C) is a **tertiary alcohol**, it **cannot be oxidized**.

(Choice A) A tertiary alcohol can be reduced to a hydrocarbon; however, the chromic acid test is an *oxidation* rather than a reduction.

(Choices C and D) An **aldehyde** can be both oxidized and reduced, but Compound C is an *alcohol* rather than an aldehyde.

Educational objective:

Oxidation of an organic molecule requires a decrease in the number of C–H bonds and an increase in the number of C–O bonds. Primary and secondary alcohols can be oxidized to carboxylic acids and ketones, respectively, by various oxidizing agents. Tertiary alcohols cannot be oxidized because the tertiary carbon does not have any C–H bonds to lose.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401925

System:
Functional Groups and Their Reactions

Students used three classification tests to study the reactivity of alcohol, aldehyde, and ketone functional groups in organic compounds. The organic compounds provided to the students are given in Table 1.

Table 1 Organic Compounds Tested

Compound	Structural Formula
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$
C	$(\text{CH}_3)_3\text{COH}$
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$

First, the students performed the Lucas test by mixing a few drops of each compound with 2 mL of the Lucas reagent (ZnCl_2 and HCl). This reagent is used to test for secondary and tertiary alcohols. A positive test result is indicated by a cloudy reaction mixture upon addition of the Lucas reagent due to the formation of an insoluble compound by an $\text{S}_{\text{N}}1$ reaction with a secondary or tertiary alcohol (Reaction 1).

R

—

OH

Reaction 1

Next, a test for alcohols and aldehydes was performed by the addition of a drop of chromic acid reagent (CrO_3 dissolved in aqueous H_2SO_4) into samples of the organic compounds. The disappearance of the orange color from chromic acid and formation of a blue-green suspension is indicative of a positive result for this test. Compound A was found to be an example of a positive result, as shown in Reaction 2.

CH_3

Reaction 2

Finally, samples of the compounds were also dissolved in water, followed by the addition of NaOH and iodine. This test is known as the iodoform test, which is used to identify methyl ketones and secondary alcohols. A positive result is indicated by the formation of iodoform (CHI_3), a yellow precipitate (Reaction 3).

R

—

C

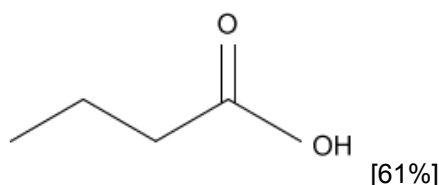
Reaction 3

Table 2 Selected Observations from the Classification Tests

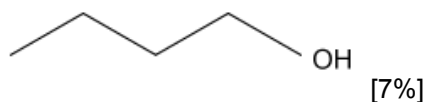
Compound	Formula	Lucas test	Chromic acid test	Iodoform test
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$	Cloudy mixture	Blue-green suspension	Yellow precipitate
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$		Blue-green suspension	No precipitate
C	$(\text{CH}_3)_3\text{COH}$	Cloudy mixture		
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$	No reaction		
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$		Orange reaction mixture	Yellow precipitate

Compound D gives a positive result for the chromic acid test. What is the structure of the product formed by the reaction of Compound D with the chromic acid reagent?

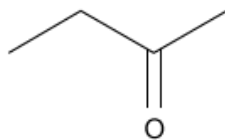
✓ ☒ A.



☐ B.

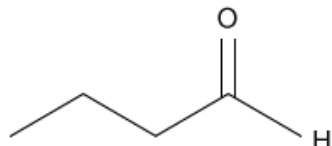


☐ C.



[11%]

☐ D.

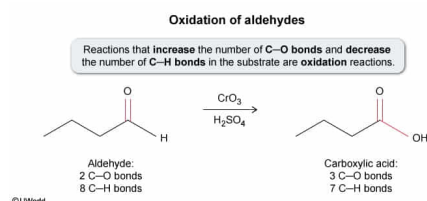


[19%]

Correct

61% answered correctly

Explanation:



Oxidation of an **organic molecule** often involves a decrease in the number of C–H bonds and an **increase** in the number of **C–O bonds**. **Oxidizing agents** will convert primary alcohols to carboxylic acids (except PCC, which converts primary alcohols to aldehydes), secondary alcohols to ketones, and **aldehydes to carboxylic acids**.

The passage demonstrates that when a secondary alcohol gives a positive result in the **chromic acid** test, it is converted to a ketone in the process. The increase in the number of C–O bonds as a result of reaction indicates that the chromic acid test is an oxidation.

Compound D is a four-carbon aldehyde (butanal), and the question states that it gives a positive result for the chromic acid test (meaning it is oxidized). Aldehydes already have two C–O bonds (as a C=O group), but when aldehydes are oxidized, they form a compound with an additional C–O bond (a carboxylic acid). Therefore, a four-carbon carboxylic acid will be produced when Compound D reacts with the chromic acid reagent.

(Choice B) Formation of 1-butanol would result from a reduction rather than an oxidation.

(Choice C) This compound is a ketone, which would result from the oxidation of a secondary alcohol rather than an aldehyde.

(Choice D) This compound is the aldehyde butanal (the same as Compound D). Because Compound D gives a positive result for the chromic acid test, it must be transformed by oxidation into a new compound rather than remain unchanged.

Educational objective:

Oxidation of organic molecules often increases the number of C–O bonds and decreases the number of C–H bonds in the molecule. Oxidizing agents, such as chromic acid, will convert primary alcohols to carboxylic acids, secondary alcohols to ketones, and aldehydes to carboxylic acids.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401926

System:
Functional Groups and Their Reactions

Students used three classification tests to study the reactivity of alcohol, aldehyde, and ketone functional groups in organic compounds. The organic compounds provided to the students are given in Table 1.

Table 1 Organic Compounds Tested

Compound	Structural Formula
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$
C	$(\text{CH}_3)_3\text{COH}$
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$

First, the students performed the Lucas test by mixing a few drops of each compound with 2 mL of the Lucas reagent (ZnCl_2 and HCl). This reagent is used to test for secondary and tertiary alcohols. A positive test result is indicated by a cloudy reaction mixture upon addition of the Lucas reagent due to the formation of an insoluble compound by an $\text{S}_{\text{N}}1$ reaction with a secondary or tertiary alcohol (Reaction 1).

R

—

OH

Reaction 1

Next, a test for alcohols and aldehydes was performed by the addition of a drop of chromic acid reagent (CrO_3 dissolved in aqueous H_2SO_4) into samples of the organic compounds. The disappearance of the orange color from chromic acid and formation of a blue-green suspension is indicative of a positive result for this test. Compound A was found to be an example of a positive result, as shown in Reaction 2.

CH_3

Reaction 2

Finally, samples of the compounds were also dissolved in water, followed by the addition of NaOH and iodine. This test is known as the iodoform test, which is used to identify methyl ketones and secondary alcohols. A positive result is indicated by the formation of iodoform (CHI_3), a yellow precipitate (Reaction 3).

R

—

C

Reaction 3

Table 2 Selected Observations from the Classification Tests

Compound	Formula	Lucas test	Chromic acid test	Iodoform test
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$	Cloudy mixture	Blue-green suspension	Yellow precipitate
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$		Blue-green suspension	No precipitate
C	$(\text{CH}_3)_3\text{COH}$	Cloudy mixture		
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$	No reaction		
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$		Orange reaction mixture	Yellow precipitate

Compound E does not give a positive result for the Lucas test. However, after reaction with LiAlH_4 , a positive result for the Lucas test is observed. Which compound is the product of the reaction of Compound E with LiAlH_4 ?

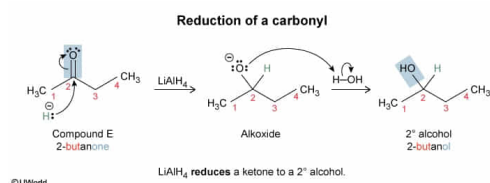
- ☐ A. Butanal [1%]
☐ B. 1-Butanol [4%]
☒ C. 2-Butanol [87%]

☐ D. 2-Butanone [5%]

Correct

87% answered correctly

Explanation:



Reduction of a carbonyl ($\text{C}=\text{O}$) decreases its oxidation state, decreases the number of **$\text{C}-\text{O}$ bonds**, and **increases** the number of **$\text{C}-\text{H}$ bonds**, yielding an **alcohol**. The metal hydride LiAlH_4 is a **reducing agent** and contains four hydride ions (H^-). The hydride ion acts as a nucleophile and attacks the electrophilic carbonyl carbon, forming a new $\text{C}-\text{H}$ bond and decreasing the number of $\text{C}-\text{O}$ bonds (a reduction). In this manner, aldehydes and **ketones** are reduced to **primary (1°) and secondary (2°) alcohols**, respectively.

Accordingly, if Compound E (a ketone) is reacted with LiAlH_4 , Compound E will be reduced to a secondary alcohol, which gives a positive result for the Lucas test. Compounds are named based on the number of carbons in the longest continuous carbon chain (**prefix**) and the **highest-priority functional group (suffix)**. The carbons are numbered to give the highest-priority functional group the smallest number, which is added to the beginning of the parent chain name.

Applying the nomenclature rules, Compound E is a four-carbon (prefix *but-*) ketone (suffix *-one*) with the name 2-butanone. The name of the 2° alcohol formed from the reduction of 2-butanone is obtained by changing the ketone suffix *-one* to the **alcohol suffix *-ol*** to give butanol. The hydroxyl group for this compound is on carbon 2, so the name of the 2° alcohol product is 2-butanol.

(Choices A and B) Reduction of a ketone results in a 2° alcohol rather than an **aldehyde** or a 1° alcohol.

(Choice D) Compound E is 2-butanone. Because it is what reacts with LiAlH_4 , it cannot be the product of the reaction.

Educational objective:

Reduction of an aldehyde or ketone with a reducing agent, such as LiAlH_4 , reduces the number of $\text{C}-\text{O}$ bonds and increases the number of $\text{C}-\text{H}$ bonds to form a 1° and 2° alcohol, respectively. Ketones are named using the suffix *-one*; alcohols are named using the suffix *-ol*.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401927

System:
Functional Groups and Their Reactions

Students used three classification tests to study the reactivity of alcohol, aldehyde, and ketone functional groups in organic compounds. The organic compounds provided to the students are given in Table 1.

Table 1 Organic Compounds Tested

Compound	Structural Formula
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$
C	$(\text{CH}_3)_3\text{COH}$
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$

First, the students performed the Lucas test by mixing a few drops of each compound with 2 mL of the Lucas reagent (ZnCl_2 and HCl). This reagent is used to test for secondary and tertiary alcohols. A positive test result is indicated by a cloudy reaction mixture upon addition of the Lucas reagent due to the formation of an insoluble compound by an $\text{S}_{\text{N}}1$ reaction with a secondary or tertiary alcohol (Reaction 1).

R

—

OI

Reaction 1

Next, a test for alcohols and aldehydes was performed by the addition of a drop of chromic acid reagent (CrO_3 dissolved in aqueous H_2SO_4) into samples of the organic compounds. The disappearance of the orange color from chromic acid and formation of a blue-green suspension is indicative of a positive result for this test. Compound A was found to be an example of a positive result, as shown in Reaction 2.

CH_3

Reaction 2

Finally, samples of the compounds were also dissolved in water, followed by the addition of NaOH and iodine. This test is known as the iodoform test, which is used to identify methyl ketones and secondary alcohols. A positive result is indicated by the formation of iodoform (CHI_3), a yellow precipitate (Reaction 3).

R

—

C

Reaction 3

Table 2 Selected Observations from the Classification Tests

Compound	Formula	Lucas test	Chromic acid test	Iodoform test
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$	Cloudy mixture	Blue-green suspension	Yellow precipitate
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$		Blue-green suspension	No precipitate
C	$(\text{CH}_3)_3\text{COH}$	Cloudy mixture		
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$	No reaction		
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$		Orange reaction mixture	Yellow precipitate

An unknown compound containing both a secondary alcohol and an aldehyde in its structure was given to a student. Suppose the Lucas test confirmed there was an alcohol present. If a protecting group were added to the alcohol, would a positive result from the chromic acid test be indicative of only the presence of the aldehyde?

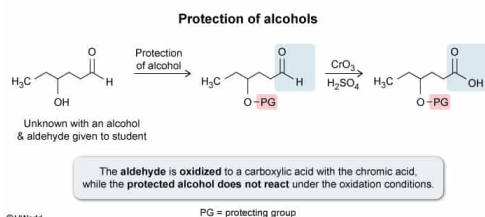
- ☐ A. Yes, because an aldehyde is unreactive under oxidizing conditions [2%]
- ☐ B. No, because an aldehyde is reactive under oxidizing conditions [8%]

- ✓ ☒ C. Yes, because a protected alcohol group is unreactive under oxidizing conditions [80%]
☐ D. No, because a protected alcohol group is reactive under oxidizing conditions [8%]

Correct

81% answered correctly

Explanation:



Organic compounds often contain multiple **functional groups** that can both react under the same conditions. In cases where a reaction is desired in one functional group but not another functional group, protecting groups are utilized. **Protecting groups** are organic molecules added to a **functional group** to prevent it from reacting and forming an undesired product. Protecting groups tend to be **stable under**

oxidizing and **reducing** conditions. After the desired reaction is complete, the protecting group can be removed to provide the original functional group.

The **chromic acid** test is an **oxidation** reaction, and the **alcohol** and **aldehyde** would both react with chromic acid to give a positive result (blue-green suspension). Primary alcohols and aldehydes will be oxidized to carboxylic acids, and secondary alcohols will be oxidized to ketones.

To confirm there is an aldehyde present in the unknown compound, the alcohol can be protected. Because **protecting groups** are **unreactive** under oxidizing conditions, only the aldehyde will be oxidized by chromic acid to form a carboxylic acid.

(Choices A and B) An aldehyde *is* reactive under oxidizing conditions, and because the chromic acid test is an oxidation reaction, an aldehyde would give a positive result. Because the protected alcohol cannot react and there are no other reactive functional groups present, a positive result is solely indicative of an aldehyde in this compound.

(Choice D) A protected alcohol is *unreactive* under oxidizing conditions, and therefore would not react in the chromic acid test.

Educational objective:

Chromic acid is an oxidizing agent that can oxidize primary alcohols and aldehydes to carboxylic acids and secondary alcohols to ketones. Protecting groups are organic molecules added to a functional group to prevent them from reacting under a given set of reaction conditions. Protecting groups tend to be stable against oxidation and reduction.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 401928
System:
Functional Groups and Their Reactions

Students used three classification tests to study the reactivity of alcohol, aldehyde, and ketone functional groups in organic compounds. The organic compounds provided to the students are given in Table 1.

Table 1 Organic Compounds Tested

Compound	Structural Formula
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$
C	$(\text{CH}_3)_3\text{COH}$
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$

First, the students performed the Lucas test by mixing a few drops of each compound with 2 mL of the Lucas reagent (ZnCl_2 and HCl). This reagent is used to test for secondary and tertiary alcohols. A positive test result is indicated by a cloudy reaction mixture upon addition of the Lucas reagent due to the formation of an insoluble compound by an $\text{S}_{\text{N}}1$ reaction with a secondary or tertiary alcohol (Reaction 1).

R

—

OH

Reaction 1

Next, a test for alcohols and aldehydes was performed by the addition of a drop of chromic acid reagent (CrO_3 dissolved in aqueous H_2SO_4) into samples of the organic compounds. The disappearance of the orange color from chromic acid and formation of a blue-green suspension is indicative of a positive result for this test. Compound A was found to be an example of a positive result, as shown in Reaction 2.

CH_3

C

Reaction 2

Finally, samples of the compounds were also dissolved in water, followed by the addition of NaOH and iodine. This test is known as the iodoform test, which is used to identify methyl ketones and secondary alcohols. A positive result is indicated by the formation of iodoform (CHI_3), a yellow precipitate (Reaction 3).

R

—

C

Reaction 3

Table 2 Selected Observations from the Classification Tests

Compound	Formula	Lucas test	Chromic acid test	Iodoform test
A	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$	Cloudy mixture	Blue-green suspension	Yellow precipitate
B	$\text{CH}_3(\text{CH}_2)_3\text{OH}$		Blue-green suspension	No precipitate
C	$(\text{CH}_3)_3\text{COH}$	Cloudy mixture		
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$	No reaction		
E	$\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$		Orange reaction mixture	Yellow precipitate

Based on the results presented in Table 2, which compounds would give a positive result for the iodoform test?

- I. $\text{CH}_3\text{CH}_2\text{CHO}$
- II. $\text{CH}_3(\text{CH}_2)_4\text{OH}$
- III. $\text{CH}_3(\text{CH}_2)_2\text{C}(\text{O})\text{CH}_3$
- IV. $\text{CH}_3(\text{CH}_2)_2\text{CH}(\text{OH})\text{CH}_3$

☐ A. I and III only [2%]

☐ B. II and III only [2%]

☐ C. II and IV only [3%]

✓ ☒ D. III and IV only [91%]

Correct

91% answered correctly

Explanation:

Iodoform test - Reaction 3



To give a positive result for the iodoform test, a compound must have an internal C–O or C=O bond adjacent to a –CH₃ for CHI₃ to form.

Compound	Functional group	Iodoform test result
I. CH_3CH_2CHO 	Aldehyde	No precipitate (negative)
II. $CH_3(CH_2)_4OH$ 	1° alcohol	No precipitate (negative)
III. $CH_3(CH_2)_2C(O)CH_3$ 	Ketone	Yellow precipitate (positive)
IV. $CH_3(CH_2)_2CH(OH)CH_3$ 	2° alcohol	Yellow precipitate (positive)

Aldehydes & 1° alcohols do not have an internal C–O bond adjacent to a –CH₃.

Ketones & 2° alcohols do have an internal C–O bond adjacent to a –CH₃.

A general expression of the iodoform test is given by Reaction 3:

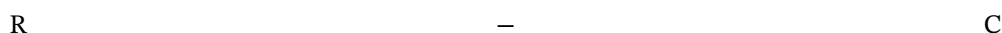


Table 2 shows that Compounds A and E give a positive result (yellow precipitate forms) whereas Compound B gives a negative result (no precipitate) for the iodoform test.

Compound A is a **secondary (2°) alcohol**, and Compound E is a methyl **ketone**. These functional groups have an internal C–O bond adjacent to a –CH₃ group, allowing for the reaction with I₂ and NaOH to occur and form the yellow precipitate. However, Compound B is a primary (1°) alcohol, which has a terminal C–O bond and therefore cannot be adjacent to a –CH₃, which prevents its reaction with I₂ and NaOH, resulting in no precipitate formation.

Among the new compounds presented in this question, CH₃(CH₂)₂C(O)CH₃ and CH₃(CH₂)₂CH(OH)CH₃ are a methyl ketone and a 2° alcohol, respectively, and each has an internal C–O bond adjacent to a –CH₃. Based on the results from the model compounds shown in Table 2, a 2° alcohol (Compound A) and a methyl ketone (Compound E) should give a positive result in the iodoform test. Because CH₃(CH₂)₂C(O)CH₃ and CH₃(CH₂)₂CH(OH)CH₃ contain the **same functional groups** as the model Compounds A and E, CH₃(CH₂)₂C(O)CH₃ and CH₃(CH₂)₂CH(OH)CH₃ (**Numbers III and IV**) will **react** in a **similar fashion** with I₂ and NaOH to give a **positive result** for the iodoform test.

(Numbers I and II) The oxygen atom in aldehydes and 1° alcohols is bonded to a terminal carbon, so these compounds are adjacent to a –CH₂– rather than a –CH₃. Because the iodoform test requires a –CH₃ adjacent to the C–O bond to form CHI₃, aldehydes and 1° alcohols do not give a positive result for the iodoform test.

Educational objective:

Analysis of the results from an organic reaction with respect to structural features of the reacting compounds is important for drawing conclusions about what types of functional groups will react with the particular reagents. Such an analysis helps predict what compounds will or will not react with the same reagent under similar conditions.